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Effect of Sn on the production of methanol during syngas conversion over Co/alumina

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ABSTRACT

Hydrocarbons are the main reaction products of syngas conversion over cobalt-based catalysts. Methanol is also produced as a minor by-product and has been proposed to be produced via a specific reaction pathway involving formates species located at the interface with the support. The poisoning of a Co/alumina catalyst with Sn was studied here and showed to induce a strong decline in methanol production, similar to that of hydrocarbons. The addition of Sn did not prevent the formation of formates as observed by *operando* diffuse reflectance FT-IR spectroscopy. However, the reactivity of formates was markedly lowered in the presence of Sn. We propose that these observations have two possible origins. Firstly, Sn addition reduced significantly the H_2 adsorption capacity of the Sn-modified samples. Hydrogen, which is required for the hydrogenation of formates to methanol, is probably activated on the cobalt metal particles, some sites of which were strongly poisoned by Sn. Secondly, Sn addition could also have specifically poisoned interfacial sites, where the most reactive formates are present.

1. Introduction

Cobalt-based catalysts typically produce linear alkanes and short alkenes during Fischer-Tropsch synthesis [1–3]. The surface of these catalysts can be conveniently investigated by monitoring the adsorption of CO by in situ and *operando* FT-IR, since CO is both a common molecular probe and a reactant in the present case. Various types of CO (ads) species are typically observed, the exact nature of the adsorption sites involved being yet unclear [4,5]. The main band of CO linearly adsorbed on metallic cobalt appears to be located in the 2030–1990 cm^{-1} range [1,6–8].

The promotion of supported cobalt by various elements such as manganese [9,10], platinum/rhenium [11], boron [12], potassium [13] and even by traces of sulfur [14] has been reported in view of modifying the activation, activity, selectivity or stability of the active phase. The promoters induced significant changes on the nature of the carbonyl present. The formation of Co^{+} species associated with a carbonyl band at 2086 cm^{-1} was proposed in the case of a S-modified sample [14].

A band at ca. 2060 cm^{-1} has been observed in some cases and assigned to CO adsorbed on a metallic phase [10] or partially oxidized cobalt atoms $Co^{\delta\delta+}$ [12,14]. A band at 2055 cm^{-1} had been observed using surface science techniques, i.e. PM-RAIRS over Co(0001), which was assigned to CO attached to unspecified defect sites [15]. We and others have proposed that these bands in the 2060–2050 cm^{-1} region could actually be related to CO adsorbed on a carbidic phase that is preferentially formed in the absence of H_2 [7,8]. The formation of surface carbide was actually noted in the case of the aforementioned surface science investigation by *ex situ* post reaction XPS analysis [15]. The kinetics of the formation of the carbidic layer would certainly depend on the adsorption parameters (e.g. pressure, temperature) and the nature of the surface probed, e.g. nanoparticles being more reactive than single crystal surfaces.

The modification of cobalt with another element is usually carried out to obtain improved catalytic activity. Yet, a poison can also be employed in the way to understand the nature of the active site. We recently demonstrated that minute concentrations of Sn poisoned cobalt and lead to the selective poisoning of sites related to multi-bonded CO [16]. These multi-bonded CO (exhibiting a C–O stretching bond vibration between ca. 2000–1700 cm^{-1}) were the main reaction intermediates in the production of hydrocarbons [16]. This conclusion could be reached based on quantitative *operando* FT-IR analyses. In contrast, linear CO(ads) located around 2030 cm^{-1} were merely spectators.

The production of alcohols is typically favoured on Cu-containing catalysts [17], yet methanol is also observed on catalysts only based on cobalt, though the methanol yields are low [8,18]. Formates have been proposed as reaction intermediates for methanol synthesis over Cu-based catalysts [19–21]. Various formate species were reported over

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$Cu/ZnAl_2O_4$ and those were proposed to exhibit different reactivity for the formation of methanol or the reverse water-gas shift reaction [22].

The surface formates observed by DRIFTS over a Co-Mg-O sample were found to be irrelevant to the formation of ethane [6]. Our group reported a two-fold reactivity of the formates present at the surface of a Siralox®-supported cobalt used for CO hydrogenation [8]. Subsequent quantitative *operando* work [18] showed that the most reactive of these formates could account for the entire methanol formed but not for the main hydrocarbons produced.

The objective of the present study was to determine any effect of tin on the activity for methanol formation of an alumina-supported cobalt catalyst. We had shown earlier that tin acted as a poison for the formation of hydrocarbons on the same material [16]. The poisoning effect was related to the titration of sites able to form multi-bonded CO(ads) at the surface of the cobalt metal nanoparticles. Whether Sn addition will also affect methanol formation is unclear, since the reaction pathways appears to be different, at least in part [18].

2. Experimental section

The preparation and physical characterization of the Co/alumina and the Sn-modified sample has been reported in details elsewhere [16]. In brief, for all samples the Co metal loading was ca. 15 wt.% and the BET surface area was ca. 110 $m^2 g^{-1}$. The metal dispersion was also similar, comprised between 7 and 10% (determined from particle size distribution obtained by TEM shown in Fig. 1). Three Sn loading were used, corresponding to Sn/Co molar ratios of 1:120, 1:60 and 1:30. High purity Ar (> 99.999%), CO (> 99%, with ca. 4000 ppm N_2 and 5000 ppm H_2) and H_2 (99.999%) gases were purchased from l'Air Liquide. The gases were used without further purification.

DRIFTS experiments were performed at ambient pressure with a high temperature DRIFT cell (from Spectra-Tech®) fitted with KBr windows, using a collector assembly. A description and properties of the cell can be found elsewhere [23]. The spectrophotometer used was a Nicolet 6700 (ThermoFischer Scientific) fitted with a liquid-N_2 cooled MCT detector. The DRIFT spectra were recorded at a resolution of 4 cm^{-1} and 8 or 32 scans accumulation depending on the time-resolution desired. The DRIFTS spectra are reported as log (1/R), where R is the sample reflectance. This pseudo-absorbance gives a better linear representation of the band intensity against surface coverage than that given by the Kubelka-Munk function for strongly absorbing media such as those based on metals supported on oxides [24]. The contribution of gas-phase CO was subtracted using a CO(g) spectrum collected at the reaction temperature over SiC, which enabled a perfect correction [25].

The catalysts were pre-reduced at 450 °C for 1 h in a flow of 55 STPmL min^{-1} of 60% H_2 in Ar. The samples were then cooled to the reaction temperature. The feed composition was: 30% CO + 60% H_2 in Ar, unless otherwise stated. The total flow rate was 30 ml min^{-1}, unless otherwise stated. The gas hourly space velocity (GHSV) was about 18,000 h^{-1}. The CO conversion was always lower than 5% at all temperatures and therefore the DRIFTS cell could be considered as a differential reactor. Note that the maximum concentration of methanol measured at the cell exit was ca. 50 ppm, while the thermodynamic equilibrium concentration corresponding to our feed at 220 °C and 1 bar is ca. 1000 ppm. The TOF of the parent Sn-free sample reported elsewhere [26] was shown to be similar to those reported in the literature.

The reactor effluent was analysed both using an Inficon quadrupolar mass spectrometer and a FT-IR gas-cell fitted in a Nicolet spectrometer. The IR gas-cell exhibited a 2 m-long pathlength and enabled measuring concentration of reaction products in the tens of ppm. Several *m/z* fragments were recorded over the mass spectrometer over the 2–80 *m/z* range.

H_2 adsorption capacity was measured using transient experiments performed in a fixed-bed reactor filled with 300 mg of catalyst diluted with SiC to reduce the dead volume in the reactor. A 4-way valve (Valco VICI) with a small internal volume (100 µl) enabled fast switching between the two feed streams, one containing Ar and the other a 10% H_2 + 10% N_2 mixture in Ar. Needle valves were used to balance pressure drop in the lines. An INFICON quadrupolar mass spectrometer was placed on-line to record the relevant *m/z* fragments. The H_2 chemisorption was carried out at 100 °C (and not at the reaction temperature of 220 °C) to favor adsorption and increase the corresponding signal and data quality. The samples were first reduced at 450 °C for 12 h in 10% H_2 in Ar, then left in Ar for 30 min at the same temperature before being brought to 100 °C in Ar.

3. Results

Typical TEM micrographs and the particle size distribution measured over the Sn-free cobalt catalyst and the three Sn-modified samples were reported elsewhere [16] but are recalled here for the sake of completeness (Fig. 1).

The *operando* DRIFTS spectra obtained at steady-state (ca. 22 h of time on stream) at 220 °C are shown in Fig. 2. The band assignment on the Sn-free sample has been discussed in details elsewhere [8,16,18] and is only briefly recalled here.

The 2926 and 2854 cm^{-1} bands were due to the stretching vibration modes of methylene groups ($-CH_2-$) of hydrocarbons and waxes adsorbed at the surface of the sample. The intensity of these bands clearly decreased with increasing content of tin (Fig. 2). This was due to the poisoning by tin of the activity for hydrocarbon formation described elsewhere [16] and recalled in Fig. 3 for methane and propene for the sake of comparison. The rate of formation of methanol also appeared to drop with increasing Sn contents (Fig. 3).

Tin poisoned sites at the surface of cobalt metal particles that were associated with multi-bonded CO absorbing at ca. 1860 cm^{-1} [16,27]. In contrast, linear CO, characterized by a band at 2031 cm^{-1} [1,7,28], were unaffected by Sn addition (Fig. 1) and were shown to be spectator species [16].

The bands in the region 1700–1300 cm^{-1} were due both to formates and carboxylates [8,16,18]. The band at ca. 1380 cm^{-1} was due to formate species, overlap of C—H bending and symmetric OCO stretching [29]. This band was not resolved in the case of the Sn-free sample due to the overlapping of a large carboxylate band located at 1460 cm^{-1} (Fig. 1). No decomposition of the spectra was attempted in this region due to the difficulty in defining a baseline.

Increased tin concentrations led to a gradual decrease of the carboxylate signal (Fig. 2, band at 1460 cm^{-1}). Carboxylates are merely CO_2 adsorbed on a basic site. The corresponding CO_2 (not measured) was likely produced through water-gas shift reaction ($CO + H_2O \rightarrow CO_2 + H_2$) involving the CO reagent and the water produced from the Fischer-Tropsch reaction ($n\ CO + 2n\ H_2 \rightarrow C_nH_{2n} + n\ H_2O$). The lower carboxylate band intensity with increasing Sn content (Fig. 2) is therefore consistent with the corresponding reduced Fischer-Tropsch activity (Fig. 3).

It is more difficult to conclude on the evolution of the concentration of formates band, even for that at 1380 cm^{-1}, because of the strong overlapping with those of carboxylates in the case of the Sn-free sample and that at Co:Sn = 120. It is yet clear that a significant concentration of formates was still present for the most highly loaded samples Sn:Co = 1/60 and 1/30 (Fig. 2).

The reactivity of these formates was assessed for each catalyst by removing CO from the CO + H_2 feed at 200 °C (the case of the carbonyl bands has been discussed in details elsewhere [16]). The difference spectra, obtained by subtracting the spectra obtained after 200 s in H_2 from those obtained at steady-state under CO + H_2, are given in Fig. 4. Interestingly, no significant formate band at 1380 cm^{-1} was noted in the cases of the most loaded Sn catalysts, meaning that the formates observed on these samples (as seen in Fig. 2) were unreactive. In contrast, the Sn-free catalyst and that with Co:Sn = 1/120 exhibited a significant decay of the formate band, much more significant than that of the carboxylate band.

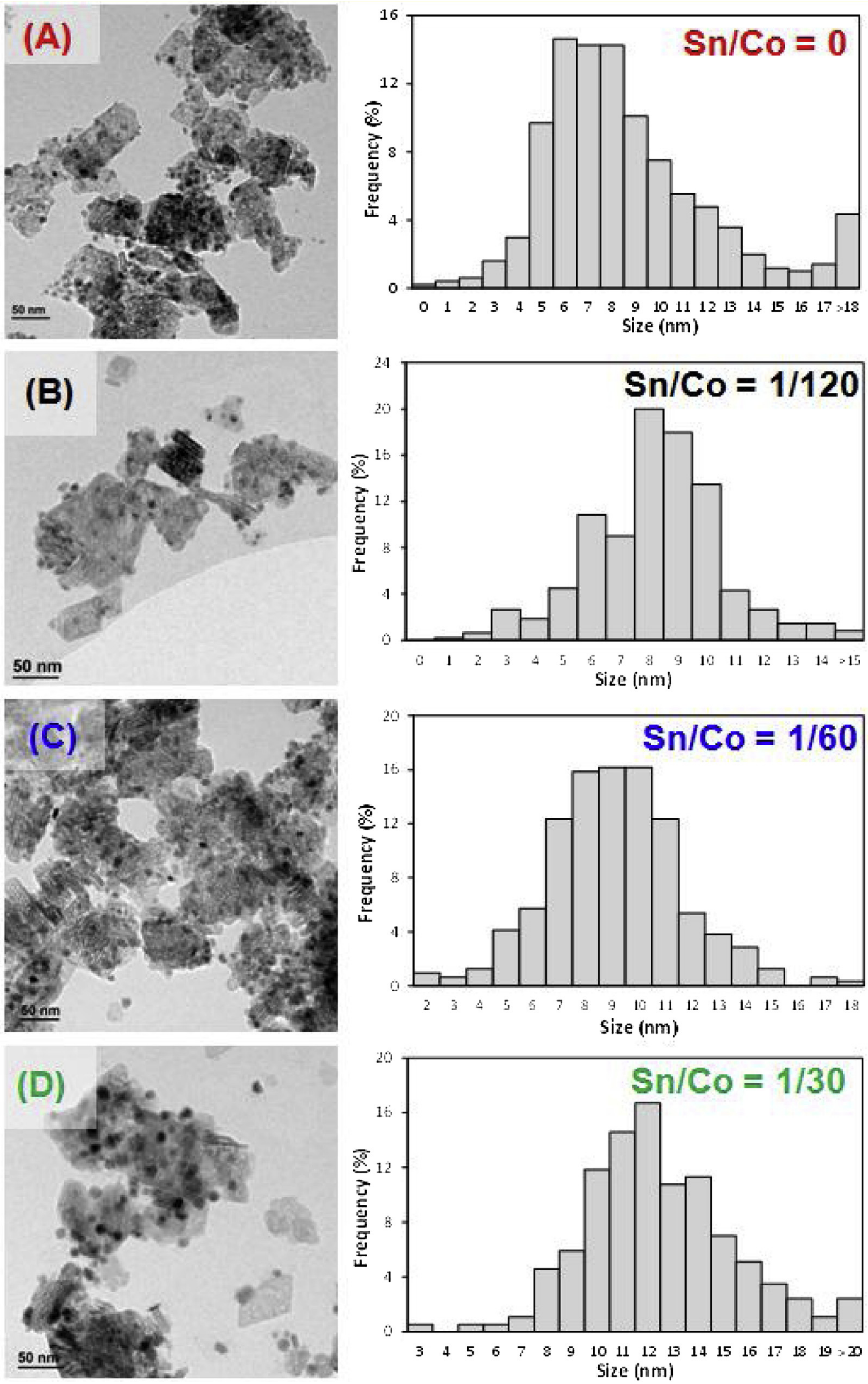

Fig. 1. Typical TEM micrograph and particle size distribution measured over the Co and Co-Sn/alumina catalysts: (A) Sn/Co = 0 (15.3 wt.% Co/alumina), (B) Sn/Co = 120 (0.24 wt.% Sn + 14.3 wt.% Co/alumina), (C) Sn/Co = 60 (0.52 wt.% Sn + 14.4 wt.% Co/alumina) and (D) Sn/Co = 30 (1.1 wt.% Sn + 14.3 wt.% Co/alumina).
(Adapted from reference 16, ©2018 Wiley-VCH).

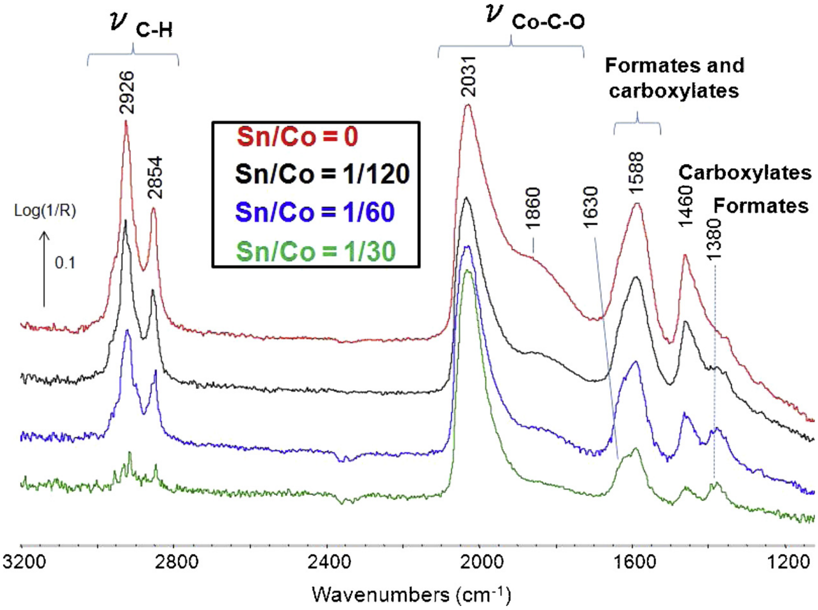

Fig. 2. *Operando* DRIFTS spectra collected at 220 °C after 22 h on stream on the Co and Co-Sn catalysts. Feed: 30% CO + 60% H_2 in Ar. The signal of the sample at 220 °C under H_2 following reduction at 450 °C was used as background. The spectrum of CO(g) was subtracted.

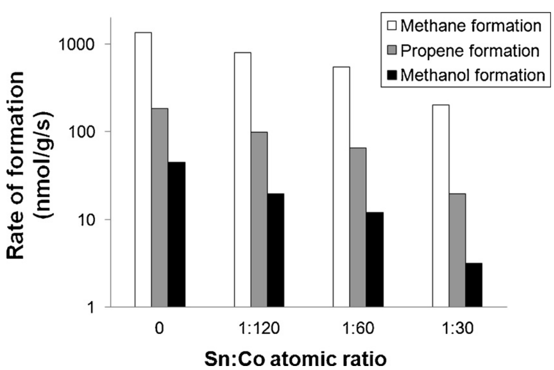

Fig. 3. Rate of formation of the main reaction products at steady-state on the Co and Co-Sn catalysts. Feed: 30% CO + 60% H_2 in Ar. T = 220 °C.

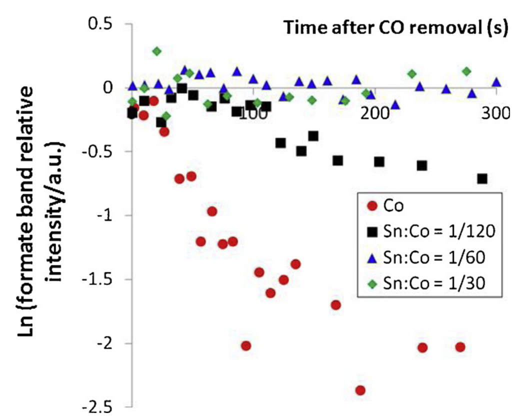

Fig. 5. Evolution with time of the formate band area at 1380 cm^{-1} at 220 °C under a H_2 stream.

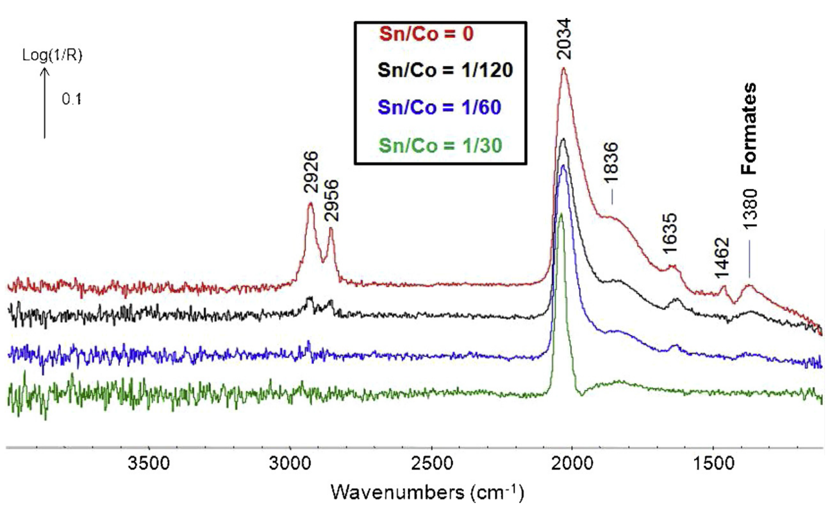

Fig. 4. Difference in situ DRIFTS spectra obtained over the Co and Co-Sn samples at 220 °C under a H_2 stream. The signal obtained after 200 s following the removal of CO was subtracted from that obtained at steady-state under CO + H_2.

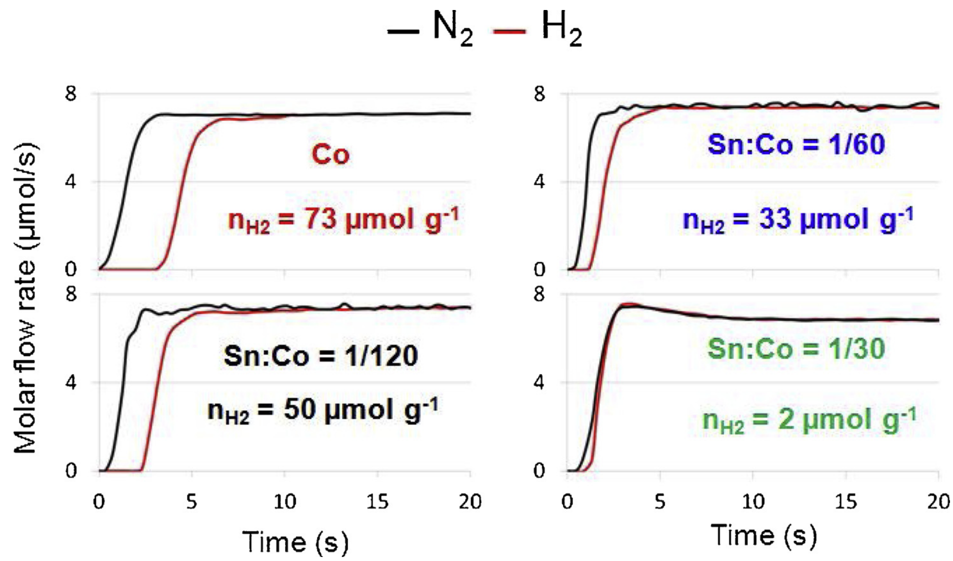

Fig. 6. Molar flow rates derived from a mass spectrometry analysis during a chemical transient experiment at 100 °C in which a flow of pure Ar was replaced with a 10% H_2 + 10% N_2 /Ar stream. The mass of catalyst was 300 mg and the total flow rate was 100 ml min^{-1}.

These trends were better evidenced when integrating the formate band in the 1400–1330 cm^{-1} region and plotting the corresponding time evolution following CO removal (Fig. 5). These plots clearly show that formates were reactive on the Sn-free sample, while the being significantly less reactive over the least Sn-loaded sample, to become mostly unreactive on the highest Sn-loaded samples over the time-scale investigated.

Surface hydroxyl groups are essential for the formation of the formate species. Unfortunately, no change in the 3700–3100 cm^{-1} region in which the surface hydroxyl groups absorb was noted following the removal of CO for any of the samples (Fig. 4). This indicates that the related concentration of OH groups involved in the formation of the reactive formates was too low to be monitored by DRIFTS.

The hydrogen adsorption capacity of the samples was determined at 100 °C by measuring the quantity of H_2 adsorbed over the reduced catalysts. The amount of H_2 adsorbed per unit of mass of catalyst was derived from the difference of H_2 molar flow rate and that of the N_2 tracer (Fig. 6). A large concentration of H_2 (i.e. 73 µmol/g) was adsorbed on the Sn-free sample. This value corresponded to about half the concentration of CO adsorbed on the same sample, i.e. 143 µmol/g [16]. This observation suggests that equivalent sites were titrated by H_2 and CO molecules, each H_2 adsorbing dissociatively on a pair of cobalt surface atoms, while CO mostly adsorbed linearly on a single atom. The data show that the H_2 adsorption capacity was then gradually reduced with increasing concentration of Sn and fell to 2 µmol/g for the sample with highest Sn content (Fig. 6).

4. Discussion

The modification of our Co/alumina catalyst with increasing concentration of tin decreased the rate of formation of methanol, similarly to the cases of methane and propene (Fig. 3). Methanol had been proposed to be formed through the intermediacy of formate species, probably through formates located at the interface with the metal [8,18,30]. Our data show that formates were still present on the Sn-loaded samples (Fig. 2) but that their reactivity was dramatically reduced as compared to the case of the Sn-free sample (Fig. 5).

Two reasons explaining this deactivation could be proposed. Firstly, Sn poisoned some interfacial sites between the cobalt metal particles and the support, in addition to the sites responsible for the formation of multi-bonded CO on the cobalt. These sites would be those normally occupied by the so-called "fast formates" that were showed to be able to account for methanol produced over a Co/Siralox® material [18].

Secondly, the lower formate reactivity could also be due a decrease in the hydrogenation ability of the Sn-modified catalysts. As a matter of fact, the H_2 adsorption capacity for the Sn-modified samples was significantly decreased with increasing Sn content (Fig. 6). H_2 uptake was essentially nil for the most Sn loaded sample, while that of CO was still significant [16]. These data indicate that sites required for H_2 adsorption and/or dissociation were poisoned by Sn, preventing further spill-over onto free cobalt sites, which remained yet available for CO adsorption. Tin exhibits a markedly lower surface tension (675 mJ/m^2) as compared to that of cobalt (2550 mJ/m^2). It is therefore likely that the Sn atoms were located at the surface of the cobalt particles, and preferably at defect sites, such as edges, corners and steps.

A plot of the methanol formation rate versus the dihydrogen adsorption capacity shows a strong correlation between these two parameters (Fig. 7). It is therefore possible that the decrease of the methanol production was primarily due to the lack of H surface species needed to hydrogenate the formate intermediate into methanol.

In the present state we have no additional data that would enable us discriminating between these two suggestions (summarized in Fig. 8), which could also occur simultaneously.

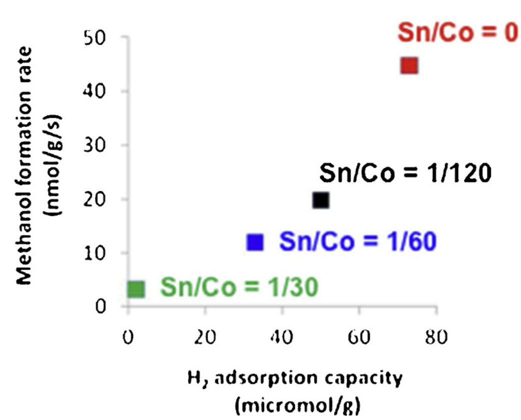

Fig. 7. Steady-state rates of formation of methanol at 220 °C under 30% CO + 60% H_2 in Ar versus the H_2 adsorption capacity for the pure Co and Sn-modified samples.

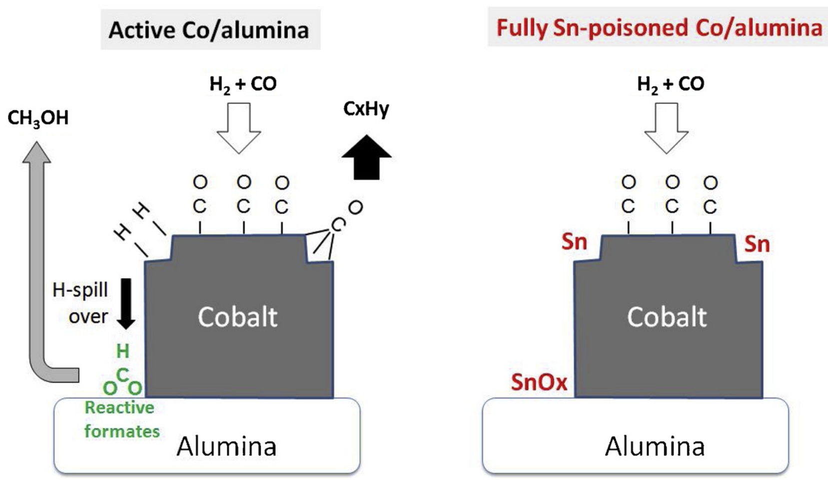

Fig. 8. Suggested reaction scheme during CO hydrogenation over cobalt supported on alumina (Left). The sites responsible for the formation of hydrocarbons are those forming multi-bonded CO (Ref. [16]). These sites get poisoned upon the addition of Sn (Right). It is suggested that these sites may also be responsible for H_2 activation, from which formates present on the support can then be hydrogenated to methanol through H spillover. The poisoning of interfacial sites between the support and cobalt metal, where reactive formates are present, can also not be excluded.

5. Conclusions

The modification of a Co/alumina catalyst with Sn was studied here and showed to induce a strong decline in methanol production, similar to that of hydrocarbons. The addition of Sn did not prevent the formation of formates as observed by operando diffuse reflectance FT-IR spectroscopy. However, the reactivity of formates was dramatically lowered in the presence of Sn. These observations are proposed to be related to (i) a reduced H_2 adsorption capacity of the Sn-modified samples, which is required for the hydrogenation of formates to methanol or (ii) the poisoning of interfacial sites, where the most reactive formates leading to methanol are located.

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